Integrated data-driven approaches to advance double-cropping recommendations for maximizing yields and soil health in the U.S. Mid-South

INTRODUCTION

Overview

With the gap between food demand and the limited potential for land expansion to achieve the projected 70% increase in food production by 2050 (Hunter et al., 2017), there is an urgent need to identify strategies for agriculture intensification (Andrade et al., 2015; Brooker et al., 2015). Alternatives to increase production without land expansion are either increasing the yields of a single crop per year per unit of land area or increasing the number of crops harvested per year (Waha et al., 2020). In the U.S. Mid-South, environmental conditions are particularly suitable for implementing double-cropping systems, which allow a second crop to be grown within the same growing season (Bryant et al., 2020). However, information is limited on how different double-cropping systems perform across environments. It is unknown the impact of different double-cropping options on long-term effects on grain yield, soil health, and water use efficiency, and this knowledge gap needs to be addressed. Our preliminary simulation results for Eastern Arkansas across 40 weather-years demonstrated that implementing a wheat-soybean double-cropping system increased year-round grain yields by 92% (soybean + wheat), protein production by 36%, and the system's water use efficiency by 42% compared to the soybean-monocrop system. To put that in perspective, for the state of Arkansas, where soybean is the number one crop with approximately 3 million acres harvested in 2023 (NASS 2023), adopting a double-cropping system on 1/3 of the soybean planted area would result in an increase of roughly 45-50 million bushels from combined soybean and wheat grain production and an added value of about $258 million to farmers, without expanding cropland area (Arkansas Agriculture Profile, 2024).

Despite the positive impact that double-cropping can have on grain production, nutritional value, and soil conservation (Crespo et al., 2023) compared to predominant monocropping systems, its adoption in Arkansas and the broader Mid-South remains limited, with less than 5% of the cropland area in more recent years (NASS, 2023). One reason is the lack of knowledge on systems' complexity, which is aggravated by unpredictable year-to-year weather variability. Without this knowledge, identifying assertive prescriptions of promising double-cropping systems for target environments remains a challenge. Thus, there is a critical need to advance prescriptions for double-cropping options, including well-established rotations such as the wheat-soybean system (Kirkpatrick et al., 2023) and new options that account for emerging winter crop markets, such as canola (*Brassica napus* L.) and oats (*Avena sativa* L.). In this proposal, our team aims to leverage an integrated data-driven approach with multi-source field, sensor, remote sensing, and crop modeling data to generate agronomic management recommendations that maximize grain production while improving soil health and water conservation, thereby supporting the sustainability and competitiveness of U.S. agriculture. This project is developed in alignment with agricultural stakeholders who recognize the urgent need to advance recommendations for double-cropped settings in the Mid-South region to maintain the competitiveness of U.S. agriculture (see the support letter from stakeholders).

Our **long-term goal** is to help stakeholders in US agriculture find paths for sustainable intensification by better understanding how production systems respond to environmental conditions. **The overall objective** of this proposal is to identify resilient soybean-based double-cropping systems that maximize yields while enhancing soil health and water use efficiency in target environments. Because double-cropping systems are complex and strongly influenced by local environmental interactions, generalized recommendations across environments often fail to identify solutions that provide a consistent yield advantage for farmers. **Our central hypothesis** is that designing site-specific recommendations for double-cropping systems using an integrative, multifaceted data-driven approach will enable a consistent increase in grain yield and enhance soil health in the U.S. Mid-South. Our combined expertise in crop physiology and modeling (Elli et al., 2025, 2022; Elli and Archontoulis, 2023), soil health, and soil conservation practices (Daniels et al., 2023; Silva et al., 2025), and remote sensing (Butts et al., 2024; Tullis et al., 2024) makes us an ideal team to lead this project. Our specific objectives are to:

Objective #1. Conduct field-scale experiments testing multiple cropping sequences to enhance understanding of soil-plant mechanisms driving crop yields and soil health. Working Hypothesis: Double-cropping systems will maximize productivity per unit land area while enhancing soil health attributes with the positive effect of the growing crop during the winter fallow period.

Objective #2: Use multispectral data at multiple scales to identify key metrics across double-cropping systems for improved biomass production and seed yields. Working Hypothesis: Integrating multispectral imagery at multiple spatial scales and machine learning algorithms will generate robust predictors of double-cropping potential including crop yields and assess crop stress, enabling optimized seed variety selection.

Objective #3. Identify double-cropping options to maximize yield levels and stability for target environments using a crop growth model (APSIM). Working Hypothesis: Site-specific double-crop recommendations will maximize crop yields and water use efficiency due to high system-environment interactions and physiological tradeoffs.

Upon the completion of this project, we expect to have established an integrated data-driven approach to generate actionable recommendations that enhance the productivity of double-cropping systems in target environments. The implementation of this framework is expected to have a positive impact on U.S. agriculture by identifying strategies for intensifying soybean-based cropping systems and generating new knowledge on prescriptions for agricultural stakeholders.

Review Of Relevant Literature

Farmers in the U.S. Mid-South have the advantage over producers in many areas of the United States of having a long growing season, which provides generous solar radiation resources and a long frost-free growing season. This unique opportunity allows them to fit a second crop into the growing season. Soybeans (*Glycine max* L. Merr), the top row crop in Arkansas with a production of 159.3 million bushels in 2023 and an estimated economic value of $2.08 billion (Arkansas Agriculture Profile, 2024), are primarily cultivated in a two-year rotation with rice (*Oryza sativa* L.), corn (*Zea mays* L.), and cotton (*Gossypium hirsutum* L.) in the U.S. Mid-South. Producers commonly select soybean maturity group 4 varieties, typically planted in April and may, with a minor portion currently cultivated in double-crop rotation with winter wheat (*Triticum aestivum*

L.). Despite its potential, the reduced adoption of double cropping (**Figure 1A**) is primarily due to the risk of delayed soybean planting, which may limit yield potential and is further complicated by the year-to-year weather variability (Ashlock et al., 2000).

A major issue of concern regarding the declining winter wheat acreage across the state of Arkansas is that there has not been an equal increase in cover crop adoption during the same period. The reduction in wheat acreage without a significant increase in cover crop adoption results in as much as 800,000 acres of land that would typically be used for growing a crop to add cash flow and prevent soil erosion that now lays fallow and is often tilled during the season with the highest rainfall, thereby leading to added soil erosion. Identifying ways to increase wheat acreage or cover crop acreage to help prevent vast tracks of land from remaining barren during the winter months will be essential to the long-term sustainability of U.S. Midsouth crop production systems. As winter wheat acreage declines in Arkansas (**Figure 1B**), alternative strategies for subsequent crops improve overall system performance by enhancing nutrient cycling, biological activity, soil organic matter, and reducing soil erosion (Wortman et al., 2012). Shifting weather patterns, such as extended frost-free periods, require rethinking traditional double-cropping systems like the wheat-soybean rotation, as well as introducing new opportunities for emerging systems.

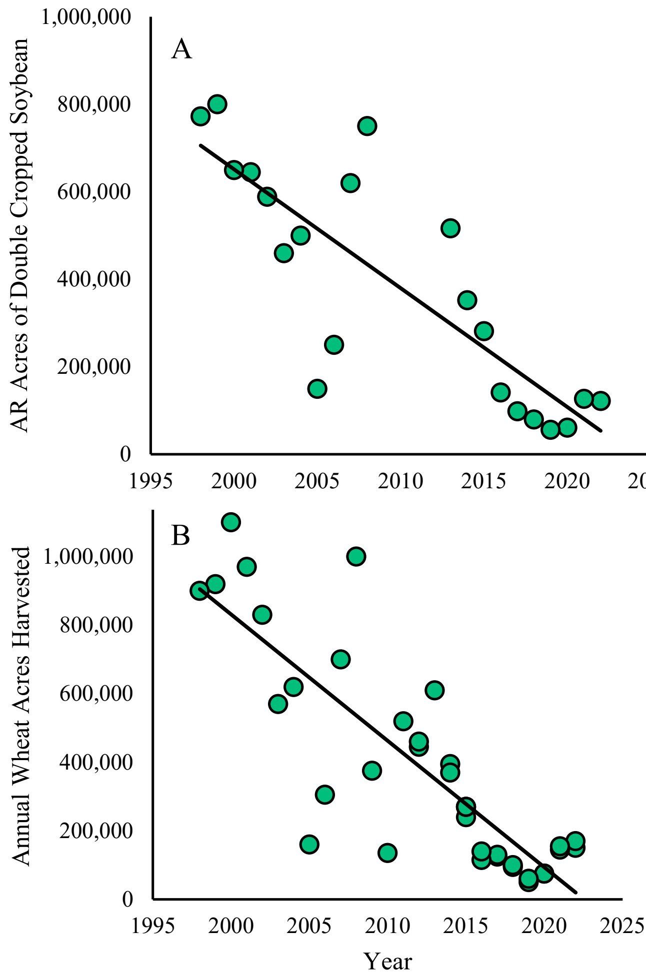

Figure 1. Historical trends in Arkansas double-cropped soybean acreage (A) and harvested wheat acreage (B) (USDA-NASS, 2023).

Given the risk of delayed soybean planting associated with wheat-soybean double cropping, we propose exploring two alternatives. The first is the soybean–wheat relay intercropping system, where soybeans are planted into a standing wheat crop before it is harvested, resulting in a period of overlap where both crops grow simultaneously (Malcomson et al., 2025). This approach has been used to maximize the yield potential of summer season crops by sowing early, while effectively suppressing weeds, optimizing land productivity, and increasing gross profit (Tanveer et al., 2017). Despite promising evidence of greater yield of soybean with land equivalent ratio (LER) value near 2.0, reduced risk of yield failure, and maximizing the net return in soybean-wheat relay cropping system, significant knowledge gaps remain in its feasibility and crop physiological responses, particularly in the U.S. Mid-South region (Yamane et al., 2016). An enhanced understanding of physiological responses when wheat and soybeans are grown concurrently could lead to the conceptualization of strategies to maximize yield. Our proposal aims to provide new insights into soybean–wheat relay intercropping feasibility under U.S. Midsouth conditions and compare this system with the traditional soybean-wheat rotation and monocropping systems. The second alternative is wheat-cotton rotation, given that cotton is less sensitive to late sowing than soybeans. While soybeans have been planted earlier in Arkansas by almost 1 day per year over the last 30 years, no substantial trends in cotton planting dates have been observed (**Figure 2**). Our proposal will provide a unique opportunity to benchmark different double-cropping systems and traditional monocropping systems for U.S. Mid-South conditions.

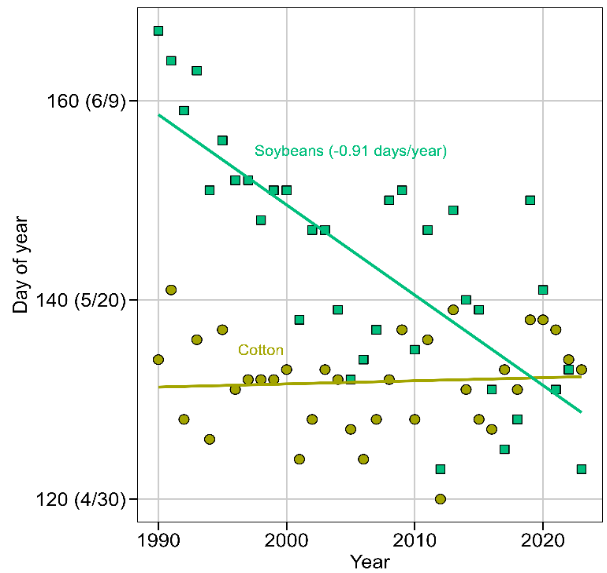

Figure 2. Historical changes in dates corresponding to USDA-NASS 50% sowing progress of soybean and cotton in Arkansas (NASS, 2023).

A previous study by Kelley and Keene (2019) in east-central Arkansas indicated that summer crop yield was significantly affected by crop sequence, demonstrating diverse variability across multiple years and cropping systems. April-planted soybeans produced higher yields when followed by corn or full-season grain sorghum compared to continuous soybean cultivation. They identified a reduced performance in corn-corn rotation and wheat planted after full-season grain sorghum. In contrast, full-season grain sorghum yields remained unaffected by the previous crop, suggesting a lower dependency on rotational effects. Overall, the inconsistent response of crop rotations to yearly environmental variability confirmed that while rotational benefits are crop- and site-specific, crop sequencing can improve yield potential and should be incorporated into crop management strategies. Double-cropped soybeans with wheat yielded lower than corn and sorghum, suggesting wheat as a preceding crop may limit soybean productivity relative to other rotations. Kirkpatrick et al. (2023) well-documented the positive effect of integrating various cover crop mixtures on soybean rotations in Arkansas. The authors found that proper planting timing and species selection improved soybean establishment by 78%. Soybean yields ranged from 1009 to 5649 kg ha^{-1}. The integration of cover crops increased soybean yield by 430 kg ha^{-1} compared to soybean following winter fallow. They identified an evident location-specific variability that requires further attention in planning cover crop mixtures, where inconsistent yields remain a significant challenge.

Arkansas row crop production is located mainly in the Mississippi Delta region, on silt loam soils, where the landscape is favorable for furrow and flood irrigation. Arkansas' frequent rainfall and warm summer temperatures associated with soil tillage practices favor increased microbial activity and crop residue decomposition, resulting in soils with very low soil organic matter (SOM) content in row crop production systems. While low SOM is not desirable due to its importance on soil chemical, physical, and biological processes that are paramount for sustaining the ecosystem, it is important to consider the singular opportunity of building SOM and increasing C sequestration with adequate soil and crop management practices. Although much of the work related to soil health has focused on the implementation of cover crops, many of the same benefits/tenets of soil health can be realized through double-cropping. There is growing interest in the agricultural industry regarding soil health and its relationship to production system resiliency.

Over the last decade, a multitude of methods to measure soil health have been developed, and soil health assessment is now a part of the services provided by private and public soil diagnostic laboratories. Soil indicators are often divided into physical, chemical, and biological categories depending on how they affect soil function, and certain soil properties such as SOM, can affect multiple soil functions or categories. There is, however, still a challenge for farmers interested in measuring soil health to choose the adequate soil health index and link the information provided in the soil-test report with practical applications. Sainju et al. (2021) highlighted that there are various approaches to measuring soil health including the Soil Management Assessment Framework (SMAF; Andrews et al., 2004; Karlen et al., 2013), Haney's Soil Health Tool (SHT; Haney et al., 2018), and Cornell University's Comprehensive Assessment of Soil Health (CASH) (Moebius-Clune et al., 2017; Fine et al., 2017). However, frequently the measurements and soil health ratings are inconclusive and do not provide consistent results due to variations in soil and climatic conditions and management practices (van Es and Karlen, 2019), which highlights the need to better investigate the applicability of these soil health assessment approaches on a regional scale, particularly in the Midsouth U.S.

With the demand for agriculture intensification and land-use efficiency, rethinking the agronomic and economic feasibility of traditional and new crop rotation options has become of urgent need. However, optimizing double-cropping sequences requires systems that acknowledge the complex interactions of growth dynamics, soil processes such as water and nitrogen cycling, and management practices. Crop growth models can help quantify and understand these interactions in response to the production systems. APSIM (Agricultural Production Systems sIMulator) is a well-established cropping system modeling framework with the capability of simulating the crop performance in diverse cropping systems, including rotations, fallow periods, and crop–environment interactions (Turpin et al., 1998; Carberry et al., 2002; Robertson et al., 2002; Verburg & Bond, 2003; Whitbread et al., 2010; Hochman et al., 2014; Keating et al., 2003; Holzworth et al., 2014). APSIM contains modules for simulating soil water balance, surface residue, soil carbon, nitrogen cycling, crop evapotranspiration (Figueiredo Moura da Silva et al., 2025; Probert et al., 1998), and the growth and development of various crops (Keating et al., 2003). Biomass production is calculated using intercepted daily solar radiation, radiation use efficiency (RUE), and environmental stressors such as water, CO_2, temperature, and nitrogen (Holzworth et al., 2014; Vanuytrecht & Thorburn, 2017).

While the APSIM framework has been extensively validated in many agroecosystems worldwide for simulating crop growth, soil water, and soil nitrogen and carbon fluxes, testing results for double-cropping systems are very limited. Zeleke and McCormick (2022) used APSIM to evaluate the performance of soybean and corn in rotation with barley, canola, faba bean, and wheat under irrigated double-cropping systems in Australia. Results indicated increased water use efficiency in summer crops. The authors concluded that, despite evidence supporting the agronomic benefits of crop rotation, several critical challenges and knowledge gaps remain that limit the development of predictive, resilient cropping systems. This confirms that in hot and dry environments, management strategies affecting yield vary based on resource availability and seasonal conditions. Chimonyo et al (2016) utilized APSIM to simulate sorghum-cowpea system performance under irrigated and rainfed environments of South Africa, capturing phenology, leaf characteristics, biomass, yield, ET, and water use efficiency. The varied yield responses to rotation across different years suggest a strong influence of interannual environmental variability.

A bottleneck in using crop growth models for predicting crop growth yield is that model testing requires extensive and expensive experimentation to generate new cultivar-specific parameters. Recent advances in high-throughput phenotyping using UAV imaging and proximal sensing can facilitate characterization of canopy development at a fine temporal scale, providing new opportunities to support calibration of crop models. Seasonal canopy coverage and remotely sensed LAI have demonstrated potential for inclusion in the crop model optimization process (Gaso et al., 2023, 2021). Publicly available time-series multispectral data from satellite platforms such as Sentinel-2 have shown promise in modeling key metrics like crop phenology, biomass accumulation, and yield potential across field to regional scales, while UAV-based multispectral imagery provides ultra-high-resolution insights into intra-field variability and localized stress responses. Additionally, by integrating these multi-scale data sources with machine learning frameworks, key vegetation indices—including NDVI, EVI, NDWI, and PRI—can be leveraged to generate accurate predictors of crop performance, detect early-season stress, and inform optimized management practices such as variety selection and irrigation timing. This multi-scale, data-driven approach supports the working hypothesis that fused remote sensing, and predictive modeling can be expanded to double-cropping evaluation and improve decision-making for sustainable intensification and expanded adoption of double-cropping practices.

Recent studies underscore the power of time-series remote sensing and machine learning for complex cropping systems. Noorazar et al. (2025) demonstrated a machine learning approach to detect and quantify double-cropping extent in areas with high crop diversity—where traditional NDVI thresholding often fails. Using Landsat-derived NDVI time series fused with expertly annotated Google Earth RGB imagery, their deep learning model achieved 96% overall accuracy, highlighting the value of multi-source, temporal data for system-level classification. Building on this, multimodal fusion has proven effective for trait prediction in individual crops under double-crop constraints. Maimaitijiang et al. (2020) used UAV-collected multispectral, thermal, and RGB imagery in soybean fields, training neural networks that achieved $R^2 = 0.72$ for yield prediction. Similarly, Da et al. (2025) fused vegetation (VI), structure (SI), and texture indices (TI) from UAV multispectral data, with gradient boosting and CNN models yielding $R^2 = 0.77$ and 0.85, respectively, for above-ground biomass. Berger et al. (2022) reviewed multi-sensor synergies in the optical domain, confirming that combining satellite and UAV platforms enhances stress detection and biomass monitoring through complementary spectral and spatial resolutions.

These recent examples support the use of multispectral data at multiple scales to identify key metrics in cropping systems and suggest that these techniques can be used to compare and evaluate different double-cropping systems. These techniques can be extended beyond single-crop method evaluation to double-cropping contexts, providing actionable tools for optimizing seed selection and management in relay and sequential systems. Our proposal aims to leverage high-dimensional phenotyping data within crop growth models to determine if model improvements can be achieved. Co-PI Davis's background in integrating Satellite/UAV and Machine Learning approaches, and PI Elli's background in crop modeling, make us well-prepared to support this proposal's tasks. The lack of studies using multifaceted frameworks that combine multiple data sources with mechanistic crop growth models under varying environmental conditions hampers progress in recommendations for double-cropping systems and their adoption. In the present study, APSIM will be used in a comprehensive set of rotations to identify short- and long-term implications on the performance of soybean-based double cropping systems in the Mid-South region to create further confidence for producers based on data-driven prescriptions. The knowledge gap we aim to fill is the limited understanding of how diverse double-cropping systems impact yields when observed from a whole-system perspective, particularly across production environments of the U.S. Mid-South.

Preliminary results

We ran a simulation exercise in east-central Arkansas using the APSIM Next Generation framework to enhance understanding of double cropping from a system's perspective. We retrieved 40 weather-years (1985-2024) from a local weather station linked to the National Weather Service network and derived soil information needed for model input from SSURGO (Soil Survey Staff, 2023). We tested a soybean monocrop system and a wheat-soybean double-cropping system. We evaluated systems outcomes on a yearly basis, starting from October 10 (winter crop sowing) through October 9 of the subsequent year. We investigated final grain yields and grain protein. Protein was estimated using values of grain yield and grain nitrogen concentration multiplied by a correction factor of 6.25, similar to the approach used by Elli et al., (2022). We determined systems' water use efficiency (WUE) for protein production, where WUE = yearly grain protein production / [yearly evapotranspiration + runoff + drainage], following the approach of Freitas et al. (2025). Preliminary results indicated that well-managed double cropping increased whole-system protein production and enhanced efficiency in converting water into protein. Seed protein production increased by 35% while WUE increased by 42% compared to the soybean-monocrop systems (**Figure 3**). Water losses (runoff + drainage) decreased by 6% under double cropping (data not shown) while systems' evapotranspiration (yearly soil evaporation + leaf transpiration) remained similar across the two systems. A detrimental impact of 7.5% on soybean yields was observed under double cropping, primarily explained by delayed sowing. This moderate detrimental impact incorporates the effect of using an early-maturing wheat cultivar, as recent warmer spring conditions have partially eliminated frost risk during wheat reproductive stages.

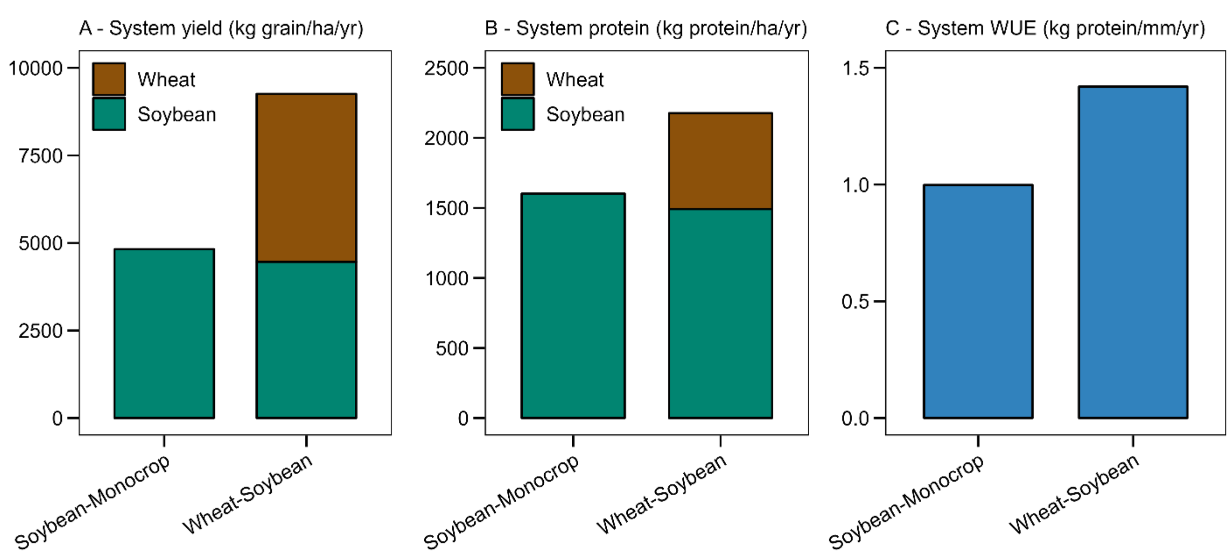

Figure 3. Systems' grain yield, protein production, and water use efficiency for protein production under soybean monocrop and wheat-soybean double cropping system in Central-East Arkansas. Values represent the average across 40 years of weather data (1985-2024).

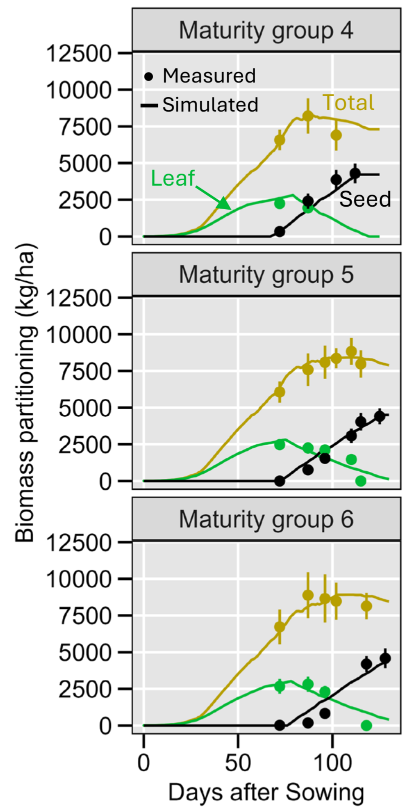

Figure 4. APSIM-Soybean simulated and measured leaf, seed, and total biomass in Fayetteville, Arkansas. Vertical lines represent the standard deviation across 4 replications.

Current efforts of the APSIM initiative are transitioning the crop models from the classic version (APSIM Classic) to a new framework, the APSIM Next Generation (APSIM NG), which accounts for improvements related to execution speed, model construction, and visualization, and manager script flexibility (Holzworth et al., 2018). While this model has proven successful in predicting growth, development, and seed yield of different crops across various environments in the U.S. (Archontoulis et al., 2020; Elli et al., 2022), we updated previous cultivar calibrations (Elli et al., 2022) to the newest APSIM NG model and further tested the model's ability to capture physiological and yield differences in soybean cultivars of varying relative maturities in the US Midsouth (**Figure 4**). While these results demonstrate the model's ability for the studied region, further testing is required across complex double-cropping systems to increase confidence in modeling predictions. We propose to use the APSIM NG model in this project for the following reasons: a) the model has proven robust in simulating several cropping systems and output variables across a range of production situations in US systems, including soil and crop N dynamics, soil water dynamics, leaf area index, phenology, and yields (Archontoulis et al., 2020; Elli et al., 2022; Pasley et al., 2020); b) the newest model version (APSIM NG) has modern languages and frameworks that allow for much greater script flexibility providing an ideal environment for extending model capabilities to simulate complex crop rotations (Brown et al., 2014); c) the model is used in more than 120 publications per year, it receives more than 3500 citations per year, and its use in research is exponentially growing based on Web of Science metrics; d) the PI has more than 7 years of experience using and improving the APSIM framework (Elli et al., 2019; Elli and Archontoulis, 2023).

Our preliminary results demonstrate: 1) the value of using a crop growth model to advance knowledge of promising double cropping systems, 2) the need to better quantify interactions genotype by environmental tradeoffs to better understand and identify promising rotations for target environments, and 3) the need to create integrate data-driven approaches to better understand the complexity of double-cropping systems and generate holistic recommendations. These findings provided the rationale for our proposed work to create an integrated, data-driven approach that utilizes multiple data types, advancing recommendations that maximize grain production per unit of land area while improving soil health and water conservation.

RATIONAL AND SIGNIFICANCE

Rationale

Enhanced understanding of complex interactions in double-cropping systems across various environments will lead to conceptualization of agronomic recommendations that maximize productivity, resource use efficiency, and soil health. Currently, farmers in the U.S. Mid-South lack actionable information on crop rotations that can maximize yields and economic returns, and our proposal aims to fill this knowledge gap.

This research addresses the program area **"Data Science for Food and Agricultural Systems (A1541)"** by using integrative new technologies and approaches to advance U.S. food and agricultural enterprises. We address intersections of two AFRI priority areas: 1) **Agriculture Systems and Technology (AST)** by developing a data-driven framework that integrates multi-source field, sensor, and remote sensing data with the APSIM crop modeling platform to generate on-farm management recommendations, and 2) **Plant Health and Production and Plant Products (PHPP)** by improving understanding of the physiological and agronomic mechanisms driving yield formation, resource-use efficiency, and stress resilience in double-cropping systems across diverse environments. These integrative approaches are expected to generate actionable recommendations that will be accessible to stakeholders beyond the project's lifetime through a web-based decision support tool.

Significance

Despite the growing interest in the U.S. in double-cropping systems for agricultural intensification, our limited understanding of the complex soil and plant interactions that affect crop growth, development, and seed yield in crop rotations is a major barrier to advancing actionable recommendations. Combining data from field, remote sensing, and crop modeling experiments can facilitate conceptualization of strategies that maximize yields while conserving soil characteristics by better quantifying the interactions of cropping systems across environments. *Our contribution is significant because new knowledge will directly inform initiatives and recommendations for farmers to maximize productivity in target environmental conditions.* Double-cropping systems provide year-round ground cover, maximizing resource use efficiency (e.g., nutrients, solar radiation, water, and land) and decreasing soil degradation and erosion compared to monoculture systems. However, the performance of cropping systems must be quantified in the context of year-to-year weather variability to develop more effective solutions to mitigate agricultural risks while enhancing both economic returns and sustaining soil quality (Moore et al., 2014; Vitale et al., 2020).

APPROACH

The proposed approach involves three major steps (**Figure 5**): 1) conduct field experiments to collect crop and soil data across multiple double-cropping systems to enhance understanding of soil-plant mechanisms driving crop yields and soil health, 2) collect remote sensing and multispectral data to identify key metrics and differences across double-cropping systems and use machine learning models to predict relevant crop traits, and 3) Use a crop growth model to identify double-cropping options that maximize yield levels and stability for target environments.

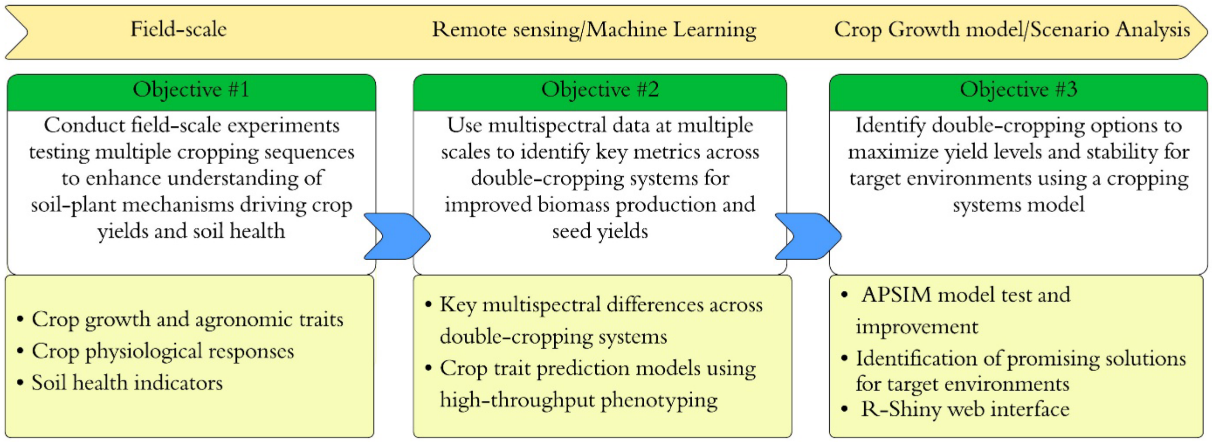

Figure 5. Objectives and main deliverables for the proposed project towards developing and applying an integrated data-driven approach to advance recommendations of double cropping systems.

Objective #1. Conduct field-scale experiments testing multiple cropping sequences to enhance understanding of soil-plant mechanisms driving crop yields and soil health

Introduction. While previous studies have shown that rotation and multi-crop sequences can enhance crop yield and soil resiliency, a comprehensive study integrating crop production, physiology, and soil health in response to different double-cropping systems is lacking. The fundamental question is how eco-physiological processes are affected by a diverse crop sequence and how they contribute to yield improvement, resource use efficiency, and soil conservation. We hypothesize that double-cropping will maximize productivity per unit land area while improving soil health attributes by implementing a growing crop during the current winter fallow period. Upon completion of this objective, we expect to elucidate the key agronomic and physiological factors that drive yield and water use efficiency across double cropping systems, and identify those crop sequences with the highest potential productivity and resilience under variable environments. The findings can be applied to develop science-based recommendations for growers and support the advancement of productive and resilience cropping systems in the U.S. Mid-South.

Task 1.1: Quantify crop growth and agronomic traits across cropping sequences. In years 1 and 2 of the project, we will conduct a field experiment with various combinations of double-cropping systems (**Table 1**) at the Pine Tree Experiment Station in Eastern Arkansas. The selected site is part of the University of Arkansas Division of Agriculture (UADA) Research & Extension Centers network and was chosen for the following reasons. First, it has all the necessary resources, equipment, and a large-scale experimental field area to conduct this experiment, as well as personnel support to help maintain the experiment. Second, PI/Co-PIs have already established long-standing, successful experiments in this research station. Lastly, Co-PI Roberts' team has a program associated with that experimental site, which will be instrumental in helping to oversee the experiment and support data collection in a timely manner. The experiment will be established using a randomized complete block design (RCBD) with four replications. Irrigation for the summer crop will be managed in accordance with recommendations adopted by the research station. Each crop rotation treatment will occupy approximately one acre, providing sufficient spatial resolution to enable reliable acquisition of satellite-based measurements (Objective 2).

Table 1. Proposed double-cropping systems that are to be investigated in this work.

#	System	Rationale
1	Continuous Soybean (years 1 & 2)	Control
2	Continuous Corn (years 1 & 2)	Control
3	Soybean (year 1) → Corn (year 2)	Control
4	Wheat (winter) → Soybean (summer)	Traditional Grain Focus
5	Wheat (winter) → Cotton (summer)	Alternative to soybean-wheat due to less cotton penalty due to late sowing
6	Wheat (winter) → Soybean relay intercropping (summer)	Emergent system with greater potential in the U.S. Mid-South
7	Oats (winter) → Soybean (summer)	Alternative winter crop/emergent market
8	Canola (winter) → Soybean (summer)	High-value winter crop/emergent market

Target planting windows and the target population are as below and will vary depending on the rotation and harvest time of the winter crop:

- **Soybean**, April 20 – June 10; 296,500–395,300 seeds/ha
- **Corn:** April 25 - April 1 – May 15; 74,100–84,000 seeds/ha
- **Cotton**: April 25 – May 15; 88,950–118,600 seeds/ha
- **Wheat:** October 15 – November 20; 3–4 million seeds/ha
- **Oat:** October 10 – November 10; 2.4–3.7 million seeds/ha
- **Canola:** October 1 – October 30; 1.5-3.0 million seeds/ha

We will collect above-ground biomass samples by hand-harvesting from a 1 m² plot area and partition them into different organs (green leaves, senesced leaves, stem, and reproductive organs) at approximately the beginning of flowering and physiological maturity of the winter and summer crops, totaling four biomass collections per year. Samples will be dried at 65 °C for 5 to 7 days and weighed accordingly. At harvest, plant height, plant density (plants per unit area), and combine yield will be recorded to evaluate management practices that optimize biomass production, seed allocation (harvest index), and final yields.

Task 1.2. Identify crop physiological responses under different cropping sequences and environmental conditions. We will utilize the LICOR 4800 system to quantify gas exchange, photosynthesis, and stomatal conductance across crop sequences throughout the growing period, providing a comprehensive picture of physiological parameters and canopy response. Measurements will span a controlled leaf temperature of 25-30 °C, a VPD (Vapor Pressure Deficit) of 1-2 kPa, a Photosynthetic Photon Flux Density (PPFD) of 1500 µmol m^{-2} s^{-1} PAR, and an average CO_2 Reference of 420 ppm. Photosynthetic rate (A_{net}), stomatal conductance (g_s), transpiration rate (E), intercellular CO_2 concentration (C_i), intrinsic water-use efficiency (WUE_i), and leaf temperature, along with chlorophyll fluorescence parameters such as F_v/F_m, ΦPSII, and NPQ, will be measured on the uppermost fully expanded, healthy, sun-exposed leaf once at the peak leaf area index (mid-season) and again during the reproductive stage to capture the physiological changes. Then, light response curves (Wu et al., 2024) will be created. Stomatal response will be combined with VPD data to assess stomatal behavior and model water use efficiency. The temperature response curve will be used to evaluate the thermal optimum of measured photosynthesis. These measurements will be supplemented by continuous measurements of soil moisture and temperature at 3 depths (15, 30, and 45 cm, covering the zone where fine roots are concentrated). The parameters and curves will link physiological characteristics to crop performance across different double-cropping systems at various times during the growing season. Intrinsic water use efficiency will be calculated using the LICOR 4800 as the ratio of net photosynthesis (A_{net}) to stomatal conductance (g_s), providing how efficiently crops fix carbon per unit of water used in stomata.

The data collected will be analyzed using statistical approaches to capture treatment effects and environmental variability. Initially, we will apply analysis of variance (ANOVA) and linear mixed-effects models (LMMs) to compare the performance of different rotation systems and quantify differences in photosynthetic traits and biomass accumulation. Correlation analyses will be conducted to examine the relationships between photosynthetic rate and stomatal conductance with biomass production and seed yield. We will fit nonlinear or spline-based models to assess biomass partitioning and growth dynamics, thereby visualizing growth patterns over time. To further understand environmental influences on crop responses, multivariate regression models with covariates, including weather, soil moisture, and temperature, will be employed using R software (R Development Core Team, 2021).

Task 1.3. Identify how soil health indicators are influenced by various double-cropping systems. Composite soil samples (n = 6) will be collected during the Spring of years 1-3 of the project. Undisturbed soil samples will be collected at a depth of 0 to 10 cm with the aid of a 5-cm inner diameter core chamber. The 0 to 10 cm depth is the current recommended soil sampling depth for routine soil analysis in Arkansas for liming and fertilization of soybean and wheat. Soil samples will also be collected from the 0- to -15 and 15- to -30 cm depth for all soil health indices. Soil samples will be placed in labeled plastic bags and soil handling (*i.e.,* oven drying, grinding, sieving, and/or maintaining a subsample at field moist condition or cooled) will be performed according to the guidelines outlined for each soil assessment or soil health index. The Soil Health Institute (SHI) suggests three key soil health indexes/measurements as being widely applied across North America (and likely beyond). Those measurements include: 1) soil organic carbon concentration; 2) carbon mineralization potential (both POX-C and soil respiration); and 3) aggregate stability (which are all included in the proposed measurements below). To evaluate the impact of the proposed treatments on soil health we will measure inherent and dynamic soil properties ($n = 24$) that are related to key soil functions (**Table 2**): *i)* sustaining biological diversity, activity, and productivity (*D*); *ii)* regulating water and solute flow (*W*); *iii)* filtering, buffering, degrading organic and inorganic materials (*F*); *iv)* storing and cycling nutrients and carbon (*N*); and *v)* providing physical stability and support (*S*) (USDA-NRCS, 2015).

The following soil properties will be measured on an annual basis due to their potential sensitivity to the proposed treatments over short time periods: pH, SOM, water extractable organic carbon, organic nitrogen and total nitrogen, H3A organic, inorganic, total P and minerals, soil respiration, enzyme activity, soil bulk density, soil compaction, Mehlich-3 nutrients and N-STaR. The following soil properties will be measured only at the beginning and end of the trial as their responsiveness to the proposed treatments may be limited on an annual time scale: soil carbon and nitrogen storage, particulate organic matter, soil water holding capacity, POX-C, and wet aggregate stability. Co-PI Roberts has conducted extensive work in soil health assessment (Daniels et al., 2023; Roberts et al., 2011, 2013; Silva et al., 2025) and is well-equipped with the necessary resources and knowledge to support this task.

Table 2. Selected soil health measurements, reference for the soil analysis procedure, and their relation to each key soil function.

Index #	Soil property	Reference/Method	Key soil function[a]				
			D	W	F	N	S
1	pH	SHT (Haney et al., 2018)	x	x	x	x	-
2	SOM		x	x	x	x	x
3	Water extractable organic C		x	-	x	x	-
4	Water extractable organic N		x	-	-	x	-
5	Water extractable total N		x	-	-	x	-
6	H3A organic P		x	-	-	x	-
7	H3A inorganic P		x	x	-	-	-
8	H3A total P		x	x	-	-	-
9	H3A minerals (Al, NH_4^+, NO_3^-, K, Ca, Mg, Na, Fe, Mn, S, Cu, and Zn)		x	x	-	-	-
10	Soil respiration (ppm CO_2)		x	-	x	x	x
11	Enzyme activity: β-glucosidase– C cycle	Dick (2011)	x	-	-	x	-
12	Enzyme activity: N-Acetyl-β-glucosaminidase – N cycle		x	-	-	x	-
13	Enzyme activity: Phosphodiesterase – P cycle		x	-	-	x	-
14	Soil bulk density	Grossman & Reinsch (2002)	x	x	-	x	x
15	Soil C storage	McKee et al. (2019)	x	x	x	x	x
16	Soil N storage		x	-	-	x	-
17	N-STaR (Potentially mineralizable N)	Roberts et al. (2011, 2013)	x	-	-	x	-
18	Mehlich-3 nutrients (P, K, Ca, Mg, Na, Fe, Mn, S, Cu, Zn, and B)	Zhang et al. (2014)	x	x	-	-	-
19	Particulate organic matter	Desrochers et al. (2019a)	x	x	x	x	x
20	Soil water holding capacity	Desrochers et al. (2019b)	-	x	x	-	-
21	POX-C	Calderon et al. (2017)	x			x	x
22	Wet aggregate stability	Fine et al. (2017)	x	x		x	x
23	Particle-size distribution (sand, clay, silt)	Gee & Or (2002)	x	x	x	x	x
24	Soil compaction via dynamic cone penetrometer	Herrick & Jones (2002)	x	x	x	x	x

[a]D = sustaining biological diversity, activity, and productivity; W = regulating water and solute flow; F = filtering, buffering and degrading organic and inorganic materials; N = storing and cycling nutrients and carbon; S = providing physical stability and support (USDA-NRCS, 2015).

Expected results from Objective 1. At the end of years 1 and 2, we will have field data on photosynthesis, stomatal conductance, biomass, yield, and WUE for all double-cropping systems.

After analyzing and interpreting the results, we expect to identify physiological mechanisms driving enhanced yields across the different systems. We aim to quantify the effects of environmental variables on biomass accumulation and its implications on soil health attributes, and to analyze the relationships between photosynthetic rate, stomatal conductance, yield, and WUE within rotations. We expect certain crops to exhibit higher yield potential and greater biomass accumulation due to their inherent species-specific traits or when grown in rotation with specific crops, thereby providing opportunities to maximize systems' performance per environment.

Potential Pitfalls and Alternative Strategies. Unanticipated weather events (e.g., storms) could compromise data collection from field experiments (though they will also enable us to measure and observe responses during these stress events). Having the experiment across two years will provide more flexibility and confidence to collect a robust dataset and implement contingency plans. The complexity within the rotational system and environment remains a key challenge in identifying the exact physiological mechanisms or characteristics that drive responses in cropping systems. Alternatively, we will use controlled environment experiments or subset field trials to validate key physiological responses observed in the main rotation trials.

Objective #2. Use multispectral data at multiple scales to identify key metrics across double-cropping systems for improved biomass production and seed yields

Introduction. Remotely sensed imagery has been demonstrated in crop metrics modeling such as estimating pre-harvest yield (Maimaitijiang et al., 2020), phenological transitions (Noorazar et al., 2025), and yield-limiting stress detection (Berger et al., 2022). The predictive potential of these data has been greatly enhanced when further explored using machine learning to integrate multimodal (multi-scale, time-series, and multispectral) inputs for improved accuracy and we suggest that these methods can be used to compare and evaluate key metrics to improve double-cropping systems.

Task 2.1. Identify key hyperspectral differences across double-cropping systems. UAV (MicaSense RedEdge-MX) and Sentinel-2 Level-2A surface reflectance imagery will be acquired on key phenological double cropping system dates and documented stress events (drought, heat, lodging) across all rotation treatments. The spatial extent of each individual field and plot unit will be collected using RTK precision for imagery cropping and VI value extraction. UAV imagery will be stitched into georeferenced, radiometrically calibrated orthomosaic band layers. NDVI, EVI, NDWI, and a modified PRI will be computed from Sentinel-2 and processed UAV imagery at 10–20 m and sub-meter resolutions, respectively, across double-crop system in Years 1–2. Time-series profiles by double cropping system will be segmented by crop phase (e.g., wheat maturity to soybean emergence) to produce a double crop system specific NDVI, EVI, NDWI, and modified PRI values for each double crop system treatment, strategic date, at each experimental site, for both years. Field and plot level VI values will be fused with relevant *in situ* data (soil texture, temporal relevant weather, stress events, harvest data, and measured biomass) for model training in subsequent tasks.

Task 2.2. Predict relevant crop traits (biomass and yield) using high-throughput phenotyping using fused satellite-UAV index time series and machine learning. We will train gradient boosting and convolutional neural network models on 70% of Year 1 and Year 2 VI data stratified by double cropping system, using measured biomass (R^2 >0.80 target) and yield (r >0.75) as response variables. We will validate models in Year 2 with holdout data (30%) with expected outputs including pre-harvest yield forecasts (±15% error), biomass accumulation curves, and variety optimization in double-crop systems by double cropping systems evaluated.

Expected results from Objective 2. We expect to collect a robust set of georeferenced multispectral imagery in Years 1 and 2 from Sentinel-2 and UAV platforms across rotations. Imagery with calculated vegetation indices (NDVI, EVI, NDWI, modified PRI) at each double cropping system scale (fields and plots) will be explored using gradient boosting, and convolutional neural network models. Predictive outputs are expected to correlate strongly ($r > 0.75$) with noted stress events, measured biomass, and harvested seed yields in double-cropping systems. The most accurate models will predict double-crop yield potential and biomass accumulation, enabling scaled variety selection and management recommendations across field-to-regional levels.

Potential Pitfalls and Alternative Strategies. Cloud cover on strategic dates remains a limitation for satellite data acquisition. However, the high temporal resolution of two Sentinel-2 platforms, together imagining the Arkansas Delta approximately every 3 days, increases opportunities for optimal clear sky captures aligned with phenological transitions or stress events. UAV multispectral imagery, collectible under overcast conditions with proper sensor calibration, further mitigates this risk of missing key data windows despite differing spatial resolutions. High-throughput phenotyping with NDVI, EVI, NDWI, and PRI has demonstrated strong correlations ($r = 0.70–0.92$) with biomass, yield, and stress detection (Da et al., 2025; Maimaitijiang et al., 2020; Berger et al., 2022). If correlations fall below $r = 0.70$ with *in situ* measurements, plot yields, or collected biomass, alternative indices (e.g., MTCI, GCI) will be evaluated.

Objective #3. Identify double-cropping options to maximize yield levels and stability for target environments using a crop growth model (APSIM)

Introduction. The APSIM framework has been extensively calibrated and applied to model both short- and long-term crop productivity and capture complex interactions between management practices and environmental conditions within farming systems (Wang et al., 2002; Carberry et al., 2009; Dimes et al., 2011, Mohanty et al., 2012). Model testing results for double-cropping systems across U.S. Mid-South conditions are limited, which adds substantial uncertainty to plant and soil simulations. This advancement will provide more accurate predictions of yield and resource-use efficiency, particularly under unpredictable climatic conditions, and support the development of more effective crop management strategies. This outcome is anticipated to enhance prescriptions of traditional and emergent double cropping systems for U.S. Mid-South conditions.

Task 3.1. Test and improve the ability of the APSIM to simulate complex cropping sequences. We will calibrate APSIM using crop growth data from Task 1.1. The APSIM framework has specific crop models and rotation routines that enable the simulation the proposed rotations. For the soybean-wheat relay cropping system, we will use the MicroClimate. (Snow and Huth, 2004) and Soil Arbitrator (Brown et al., 2019) Modules to account for resource sharing when two crops grow together (Githui et al., 2023). We will retrieve weather information from satellite-based products such as NASA/POWER (Stackhouse et al., 2015) and CHIRPS (Funk et al., 2015), which have proven robust for use in crop modeling (Battisti et al., 2019; Luetkemeier et al., 2018). Soil inputs will be obtained by combining SSURGO (Soil Survey Staff, 2023). SOM, bulk density, and soil water holding capacity information (task 1.3) will also be used to develop the APSIM soil profile. Seasonal canopy coverage retrieved from UAV platforms (Objective 2) will be incorporated into the model calibration and testing processes to determine whether improvements in prediction are realized (Gaso et al., 2023, 2021).

Model calibration will consist of manually changing cultivar parameters within a realistic range until a suitable fit is obtained, similar to the approaches used by Elli et al. (2020a) and Irmak et al. (2024). Briefly, the first step will be adjusting phenology-related parameters to determine growing degree days and sensitivity to photoperiod so that simulated dates of flowering and maturity are within ±15% of the observations. A second step will be adjusting other model parameters related to leaf area and biomass partitioning within plausible ranges (±20%) to achieve a target normalized Root Mean Square Error – nRMSE of 20% or lower for final yields. Additional model performance metrics include coefficient of Determination – R^2 (target criteria is $R^2 \geq 0.8$ or greater for final seed yields), Willmott Agreement Index - d (Willmott et al., 1985, target criteria is d ≥ 0.8), and Modelling Efficiency Index – E (Nash and Sutcliffe, 1970, target criteria is E 0.75).

Task 3.2. Identify the most effective double-cropping system to maximize yield levels and stability for target environments using the APSIM framework. We will run APSIM simulations continuously over 40 historical years of weather data (1985-2024) to account for year-to-year weather variability. Our simulation experiment will include 10 environments, mostly located in the Eastern Delta region (**Figure 6**). The selected environments are strategically spread across the state to cover a representative environmental gradient and relevant soybean production areas. We will retrieve weather information from NASA/POWER (Stackhouse et al., 2015) and CHIRPS (Funk et al., 2015), and soil information from SSURGO (Soil Survey Staff, 2023). A 10-year spin-up period will be run to reduce uncertainties regarding initial soil conditions (Dietzel et al., 2016). A continuous simulation will be run without a yearly reset, starting January 1st, 1985, until December 31st, 2024, to account for carryover effects in the double-cropping systems (Freitas et al., 2025).

The outputs and metrics to evaluate systems performance will be yield levels and stability, systems WUE for yield (Dietzel et al., 2016, yWUE = seed yields / [evapotranspiration + runoff + drainage]), and protein (Freitas et al., 2025, pWUE = protein yields / [evapotranspiration + runoff + drainage]), cumulative intercepted solar radiation (Edwards et al., 2005), systems evapotranspiration, soil moisture at 15,30 and 45 cm, daily crop growth rates. Weather conditions will be characterized by grouping weather years into five classes based on season, temperature, and rainfall: cool/wet, warm/wet, cool/dry, warm/dry, and "normal" conditions. "Normal" conditions include years when weather conditions are within ±0.5 * standard deviation of the long-term average values (Ruane and McDermid, 2017). The dataset generated from the scenario analysis will be analyzed using R (R Development Core Team, 2021), where we will identify the optimal solution that maximizes system-level yield and stability, as well as water use efficiency per environment.

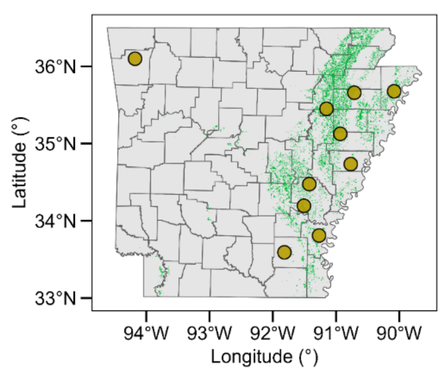

Figure 6. Arkansas map with the locations that will be used for the scenario analysis. The green background color denotes the 2023 soybean crop layer.

Task 3.3. Develop an R-Shiny web interface to maximize the dissemination of our research findings. An R-Shiny application (Chang et al., 2024) will be developed to create a user-friendly web interface that allows users to visualize results with optimal management and physiological solutions. Our goal with the web interface is to facilitate communication of the results with stakeholders, including private and industrial agriculture sectors, government organizations, and academics. We will add results from the scenario analysis to help users understand the implications of different double-cropping systems and sites on yields and sustainability metrics according to weather scenarios. Outcomes from historical weather simulations (1985-2024) will inform actionable decision-making, enabling stakeholders to adopt them rapidly. The R-shiny application will be web-based, ensuring accessibility through any standard web browser. Documentation will be provided to facilitate the use and interpretation of results.

Expected results from Objective 3. We expect to develop new cultivar-specific parameters within the APSIM framework, thereby enhancing its capacity to simulate growth, biomass accumulation, and yield outcomes across double-cropping systems. The scenario analysis will generate a dataset that includes the rotations, environments, years, and output metrics (e.g., yields, water use efficiency, evapotranspiration). We expect to identify crop sequences that maximize yields and water-use efficiency, thereby providing region-specific, science-based decision-support tools to guide farmers in making the most informed decisions for their target environments.

Potential Pitfalls and Alternative Strategies. The complex interactions between physiological output assessment, soil condition, nutrient cycling, and residual properties from previous crops present a significant challenge for isolating specific properties. Alternative strategies to address this gap will be conducting a one-at-a-time sensitivity analysis with weather, crop, and soil model parameters to enhance understanding of the system's tradeoffs and synergies. What-if scenarios will generate a large number of simulations and a large dataset, which may require high computational power to create and analyze the results. If needed, the Arkansas High-Performance Computing Center (AHPCC) provides high-performance computing hardware, storage, and support services, including training and education, to enable computationally intensive research at the university and within the state, as well as with collaborators elsewhere.

Sustainability and data management plan

Our proposal data management is **committed to following the FAIR (Findable, Accessible, Interoperable, and Reusable) standards**. All data generated, including crop and soil field measurements, sensor data, remote sensing metrics, and crop growth model outputs, will be deposited in publicly accessible, version-controlled repositories such as Ag Data Commons and GitHub with detailed metadata and permanent Digital Object Identifiers (DOIs). Codes for data analysis will be version-controlled and publicly archived on GitHub to ensure data transparency and reproducibility. Our proposed data management plan, along with the web-based decision support tool (task 3.3), aims to ensure <u>long-term data accessibility, reproducibility, and the project's lasting impact beyond its award period</u>. We will work closely with our stakeholder advisory panel (see support letter) to ensure the project's deliverables align with the needs of growers and that the decision support tool generates actionable recommendations for double-cropping systems. PI/Co-PIs will review the data management plan every 6 months to ensure that data transparency and reproducibility are adequate and make any necessary adjustments.

Dissemination of Results and Future Directions

We will disseminate our findings through peer-reviewed papers, scientific conferences, and extension events, such as field days. We will also communicate the results to farmers, industry, and policymakers through the Agricultural Communications Services, in collaboration with the Arkansas Agricultural Experiment Station. The R-Shiny web interface will be used to maximize the dissemination of our research findings to a wide range of stakeholders. PI Elli will disseminate results from this proposal into his graduate-level Crop Physiology course and undergraduate-level Soybean Production course. Students will use the proposed web-based tool to interact with the generated data, visualize prescriptions that maximize crop yields and water use efficiency. Our crop modeling scenario analysis will consider 10 environments in the US Mid-South. Performing simulations across additional regions of the U.S. is a logical next step, but it is beyond the scope of this proposal.

Project Timetable

Study Objectives and Benchmarks	**Year 1**				**Year 2**				**Year 3**			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Objective #1: Conduct field-scale experiments testing multiple cropping sequences to enhance understanding of soil-plant mechanisms driving crop yields and soil health												
Task 1.1: Quantify crop growth and agronomic traits		■	■	■	■	■	■	■				
Task 1.2: Identify crop physiological responses		■	■	■	■	■	■	■				
Task 1.3: Identify how soil health indicators			■	■			■	■			■	■
Objective #2: Use hyperspectral data at multiple scales to identify key metrics across double-cropping systems for improved biomass production and seed yields												
Task 2.1: Identify key hyperspectral differences		■	■	■	■	■	■	■				
Task 2.2: Predict relevant crop traits		■	■	■	■	■	■	■				
Objective #3: Identify double-cropping options to maximize yield levels and stability for target environments using a crop growth model (APSIM)												
Task 3.1: Test and improve the ability of the APSIM model			■	■	■	■	■	■				
Task. 3.2: Identify effective cropping systems per environment					■	■	■	■	■	■	■	■
Task 3.3: Develop an R-Shiny web interface									■	■	■	■